BUILDING A SCALABLE RECOMMENDATION SYSTEM USING AMAZON REVIEWS DATA

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PROJECT OVERVIEW

Pipeline:

Amazon Review Dataset
↓
Data Cleaning
↓
Task 1: Database Systems
↓
Task 2: Vector Similarity
↓
Task 3: Clustering
↓
Task 4: Recommendation System

AMAZON REVIEW DATA SET 2023

- 571 million reviews
- 33 product categories
- All_Beauty subset
- Reviews + Product metadata

Product Feature	Datatype
Column_Name	Data_Type
main_category	str
title	str
average_rating	float64
rating_number	int64
features	object
description	object
price	float64
images	object
videos	object
store	str
categories	object
details	object
parent_asin	str
bought_together	float64

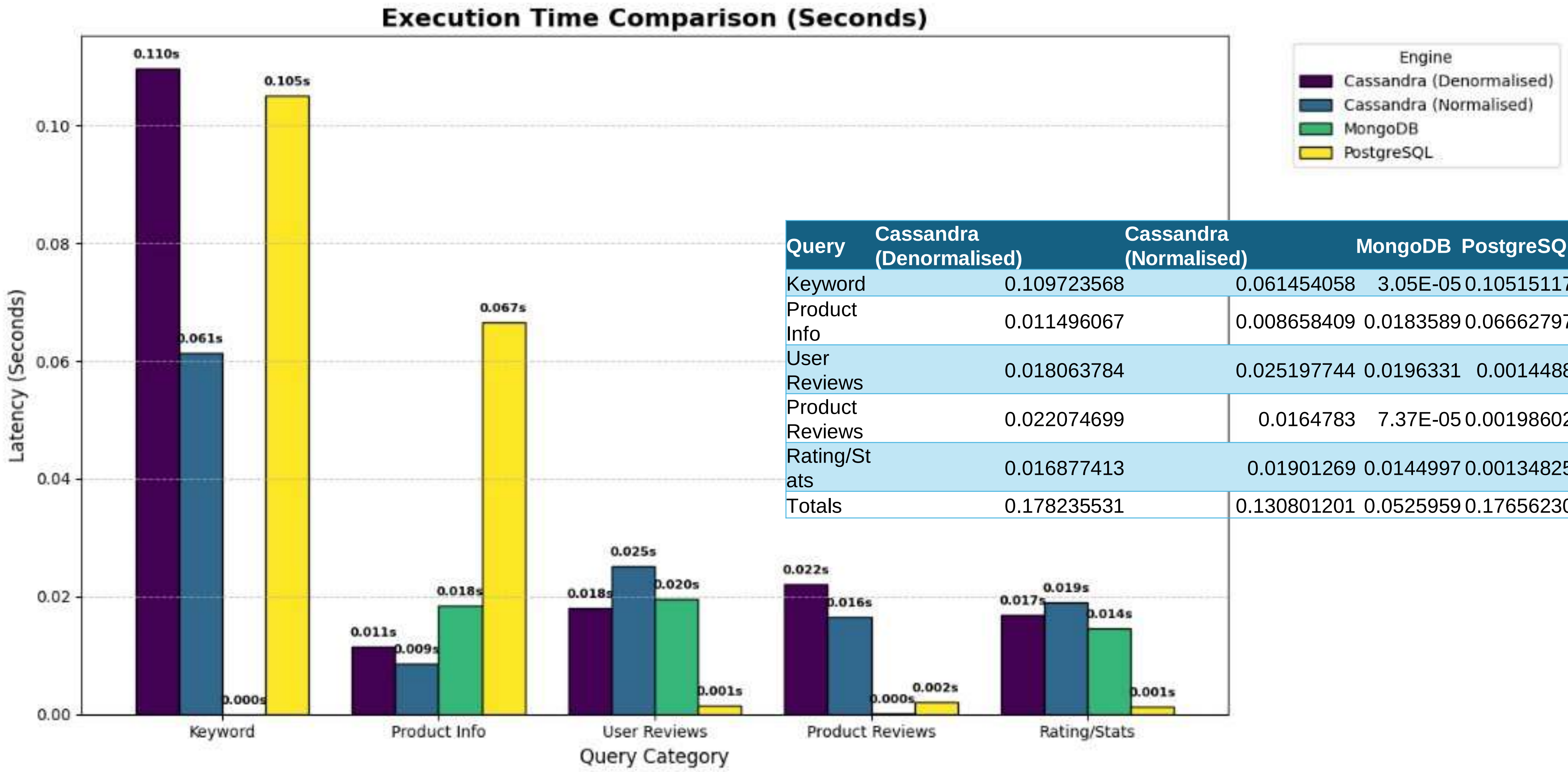
User Feature	Datatype
Column_Name	Data_Type
rating	int64
title	str
text	str
images	object
asin	str
parent_asin	str
user_id	str
timestamp	datetime64[ms]
helpful_vote	int64
verified_purchase	bool

DATA PRE-PROCESSING

Pipeline:

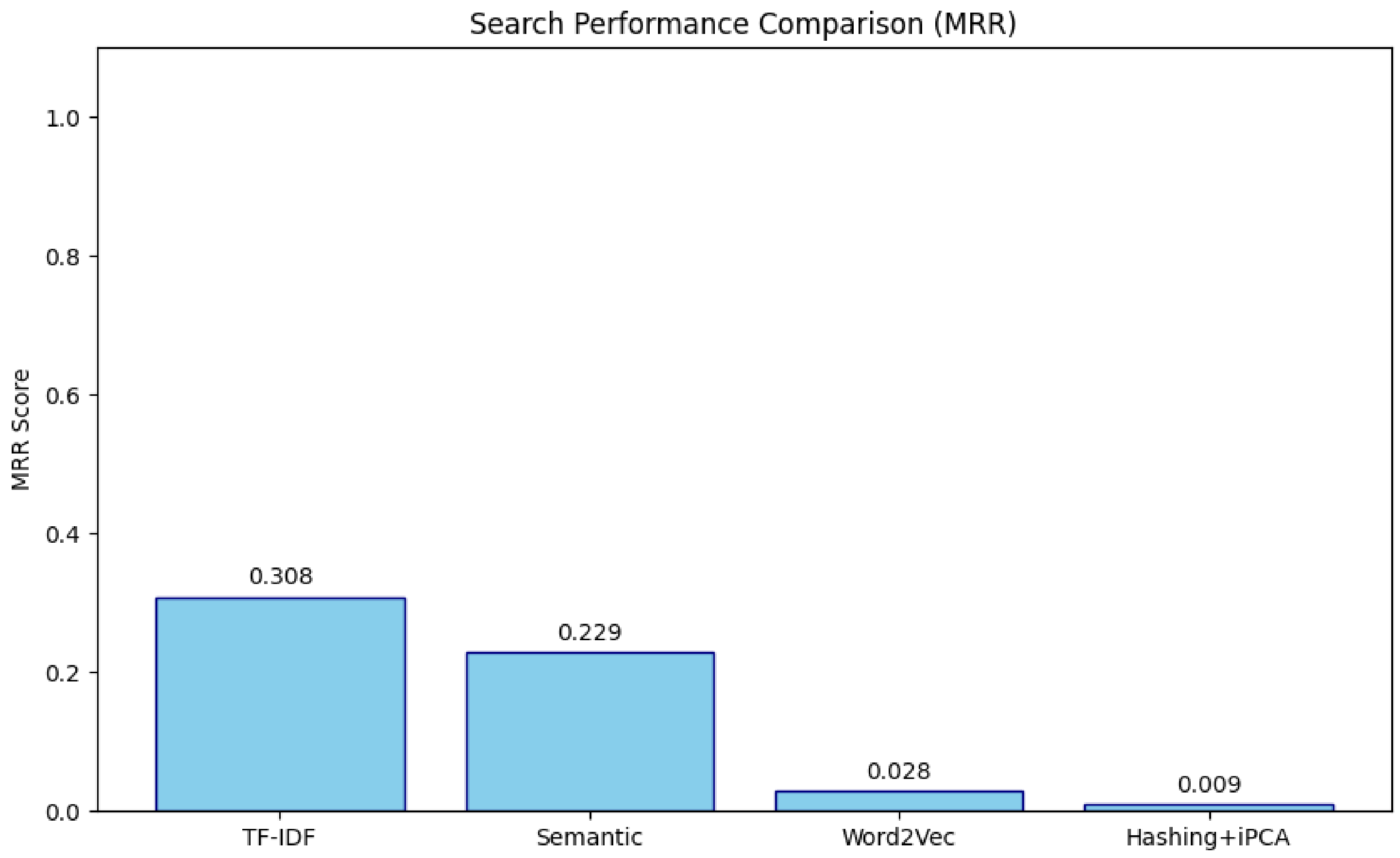
Convert text to string and remove html and links
↓
Convert Emoji to their meaning
↓
Normalise to lower case
↓
Tokenisation and remove stop words
↓
Lemmatisation and fix elongated words
↓
Join tokens back to single string

TASK 1: DATABASE SYSTEM COMPARISON



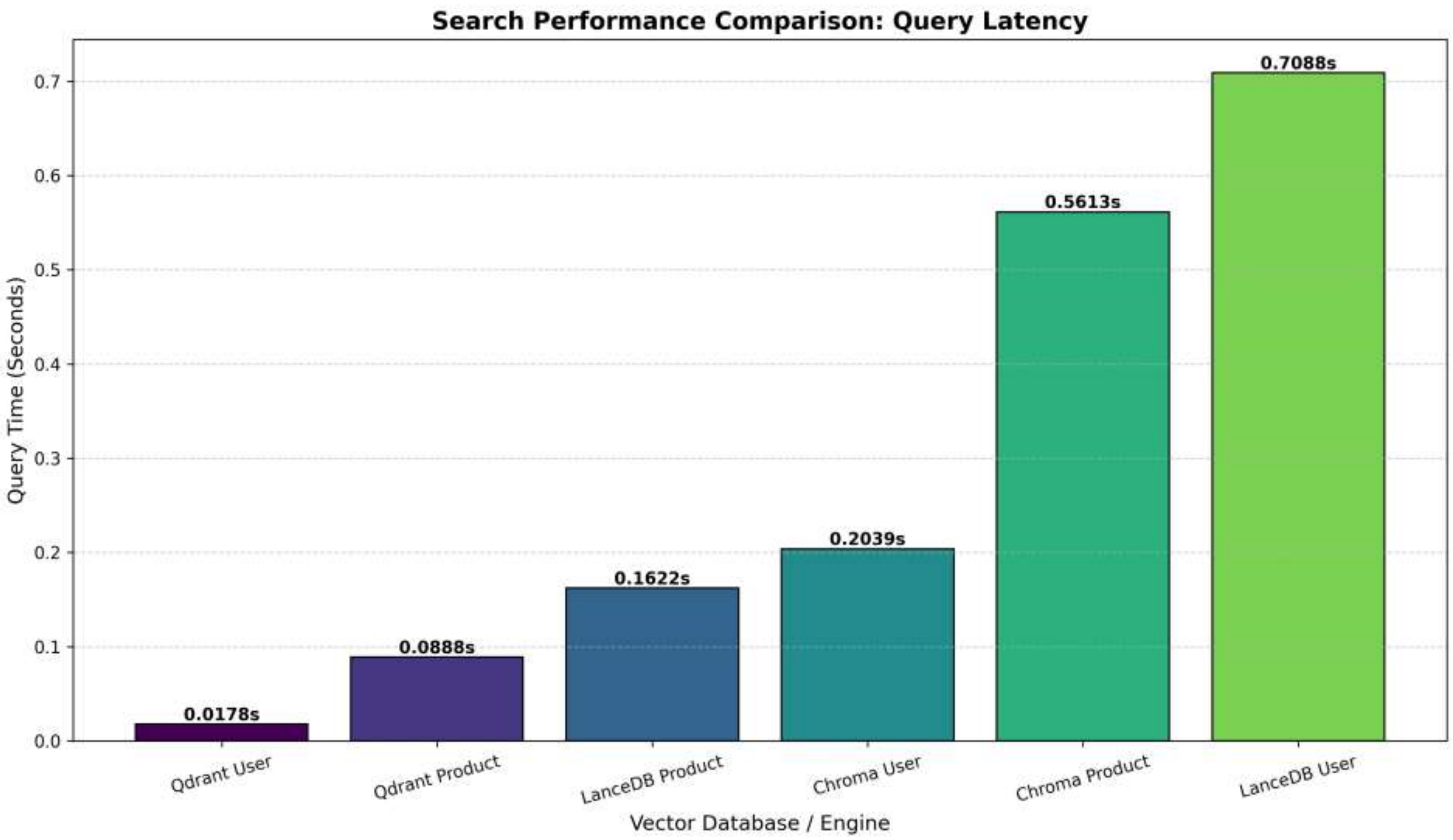
Query	Cassandra (Denormalised)	Cassandra (Normalised)	MongoDB	PostgreSQL
Keyword	0.109723568	0.061454058	3.05E-05	0.105151176
Product Info	0.011496067	0.008658409	0.0183589	0.066627979
User Reviews	0.018063784	0.025197744	0.0196331	0.00144887
Product Reviews	0.022074699	0.0164783	7.37E-05	0.001986027
Rating/Stats	0.016877413	0.01901269	0.0144997	0.001348257
Totals	0.178235531	0.130801201	0.0525959	0.176562309

VECTORISATION



Method	MRR
TF-IDF	0.308333
Semantic	0.229167
Word2Vec	0.027778
Hashing+iPCA	0.009259

TASK 2: VECTOR DATABASE COMPARISON



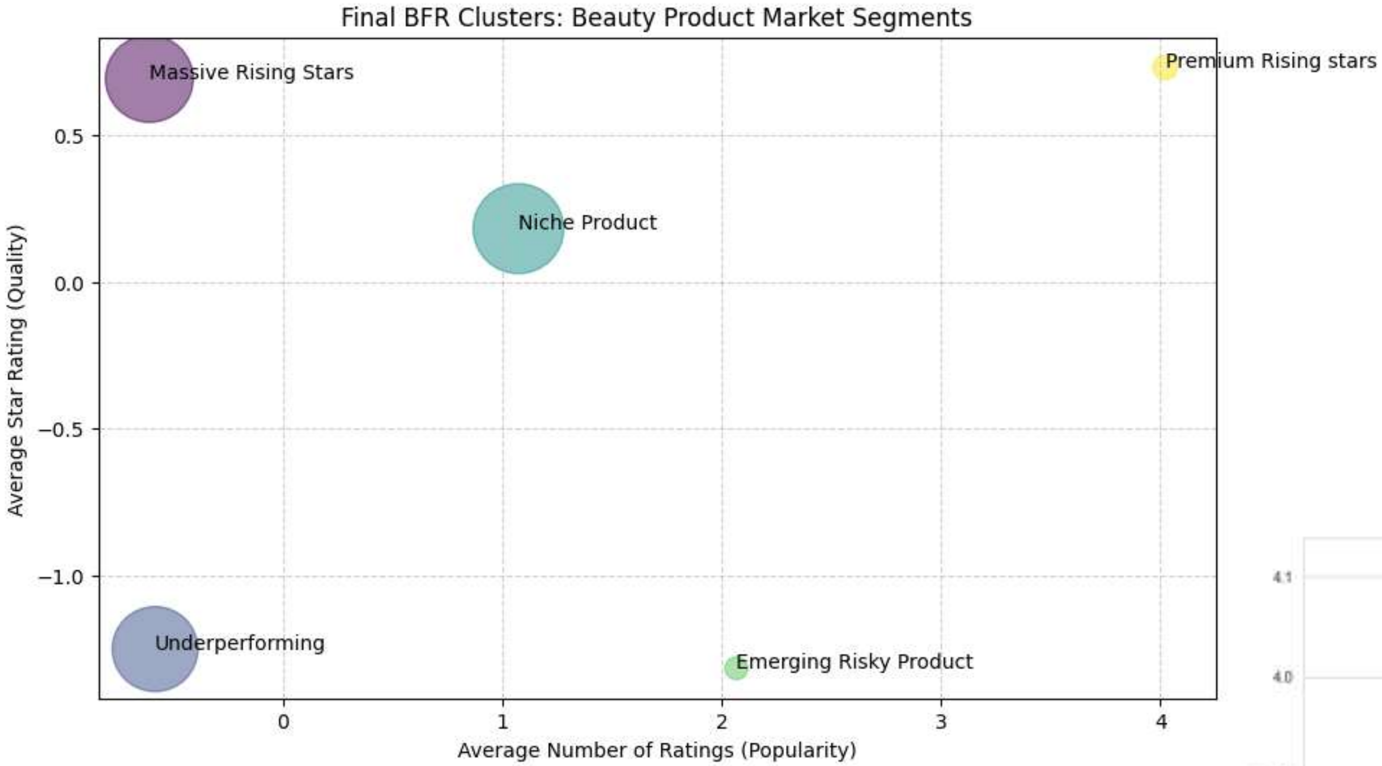
Engine	Time (Seconds)
Qdrant User	0.01781559
Qdrant Product	0.08884263
LanceDB Product	0.162170887
Chroma User	0.20390892
Chroma Product	0.561287403
LanceDB User	0.708806276

TASK 2: VECTOR SIMILARITY SEARCH

Rank - Product	ASIN	Product Title	Similarity
1	B07QWBWKVW	wide tooth comb curly hair nat..	0.927
2	B07BJ492JM	wood comb wooden hair comb 100..	0.921
3	B07BF4T557	wood comb wooden hair comb 100..	0.921
4	B07WPL2LZJ	wood comb anti static natural ..	0.908
5	B07S7P8NJ5	xuanli hair comb detangling wi..	0.901
6	B07PPBJFWC	xuanli hair comb detangling wi..	0.901
7	B079VHXTXP	wood comb wooden hair comb 100..	0.9
8	B07QX9XTKR	natural buffalo horn comb ruby..	0.899
9	B0778RTNWT	thee natural horn comb anti st..	0.892
10	B0778HW8QF	thee natural horn comb anti st..	0.892

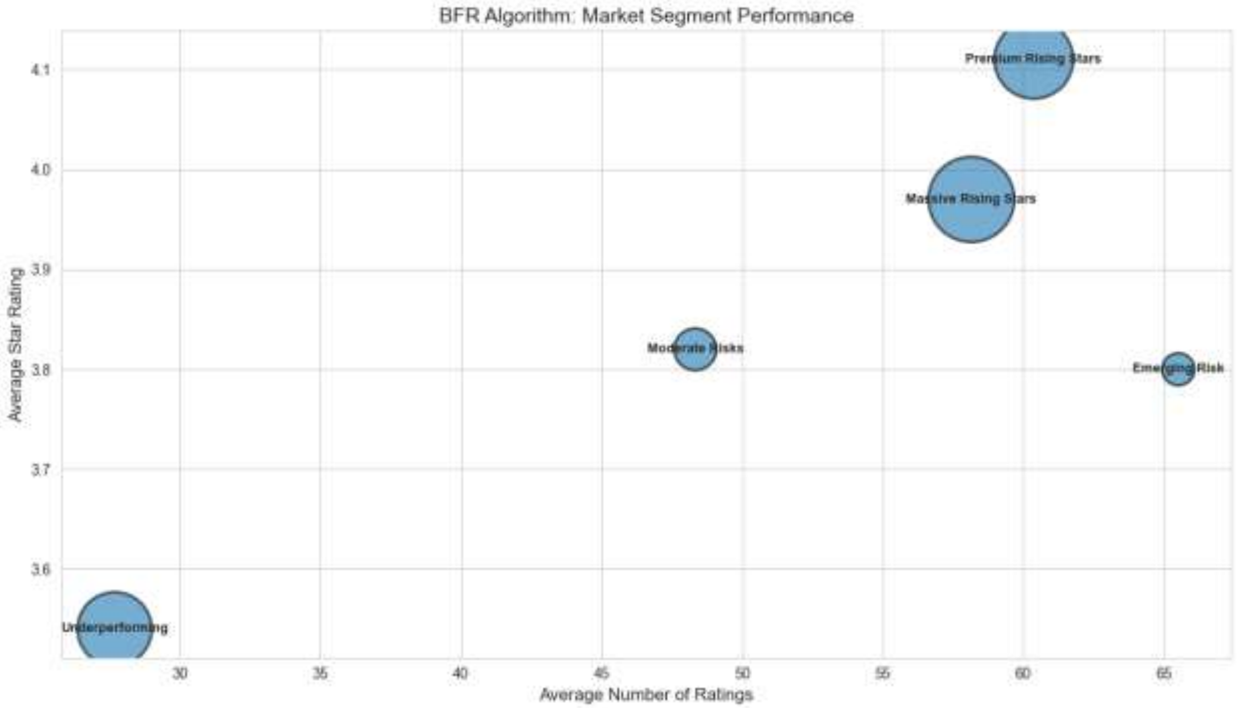
Rank - User	User ID	Similarity
1	AHV6QCNBJNSGLATP56JAWJ3C4G2A	0.706
2	AGPVUWYFDS4WHWJJC4336KQUTAPQ	0.678
3	AFS5MAXWHQNKTORY7CNERRHNV7DA	0.671
4	AEDRFBA64VH2SHGAB5W27J4ZTKHA	0.667
5	AFKSNZYMIZ6WAU3A7RREIJ3RIABA	0.665
6	AHY6N2ZTVLFE7N5OWEPCG2C3ODTQ	0.665
7	AHSMHVELSPKUCSUPDPNRJE2JOIFQ	0.665
8	AF7SECCKJFGECV6A7YJEZL2NWFXA	0.663
9	AHYCC2Y7SQ6QRFSUKDTAUWWAH2FA	0.662
10	AGQVY4TAXPWFHDCKVXJG5KMXBKFA	0.662

TASK 3 :PRODUCT CLUSTERING



Business insights

- Improve product quality for risky products
- Use viral products for marketing



TASK 3 : REVIEW CLUSTERING

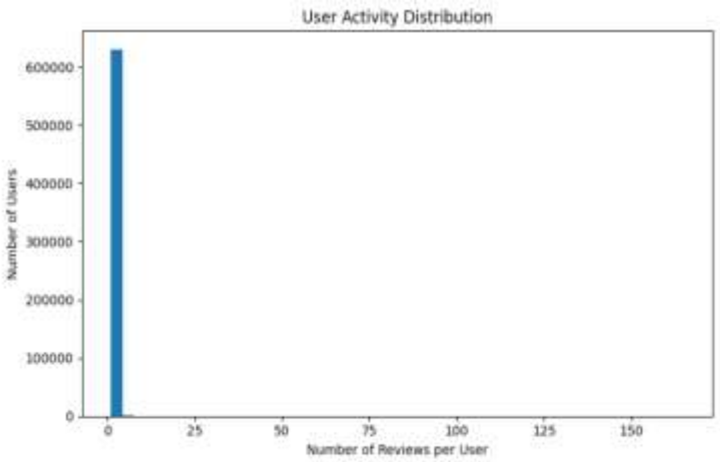
Cluster Name	Sentiment Score	Review Length
Premium Negative Review	-0.59	387.92
Premium Positive Review	0.85	453.33
Standard Negative Review	-0.7	70.49
Stand Positive Review	0.78	74.48

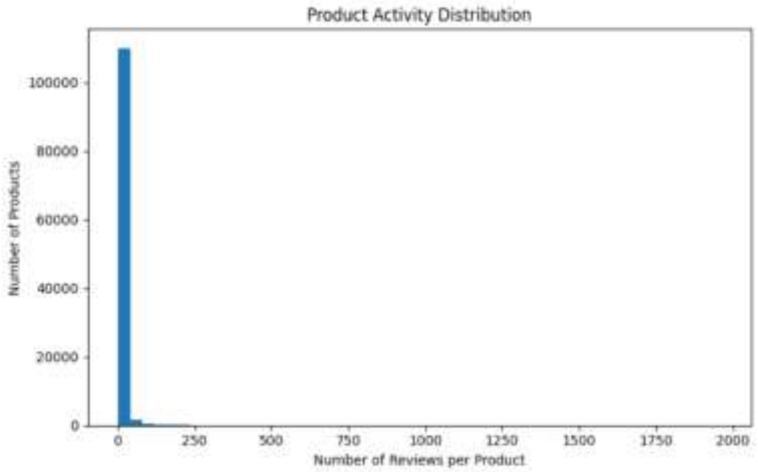
Cluster Name	Cluster Size	Average Recency	Seasonal Profile
Summer customer	143621	1,676 days	100% Summer
Winter customer	148763	1,708 days	100% Winter
Spring customer	153610	1,730 days	100% Spring
Autumn customer	138525	1,729 days	100% Autumn
Long-Recency Winter customer	12060	2,200 days	100% Winter
Summer customer	35407	1,673 days	33% Summer

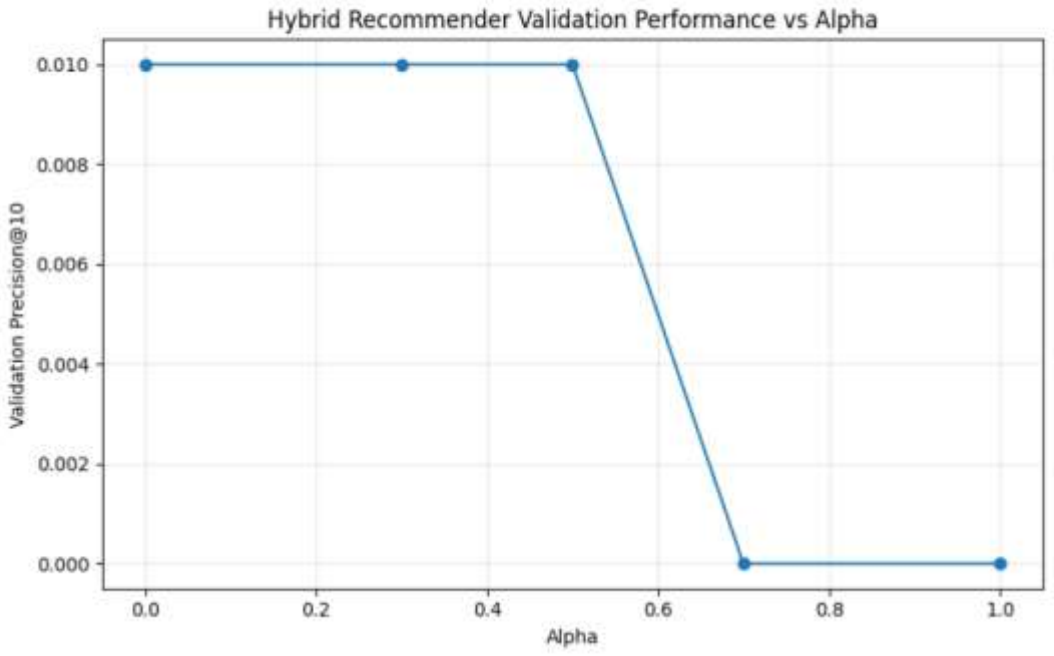
Business insights

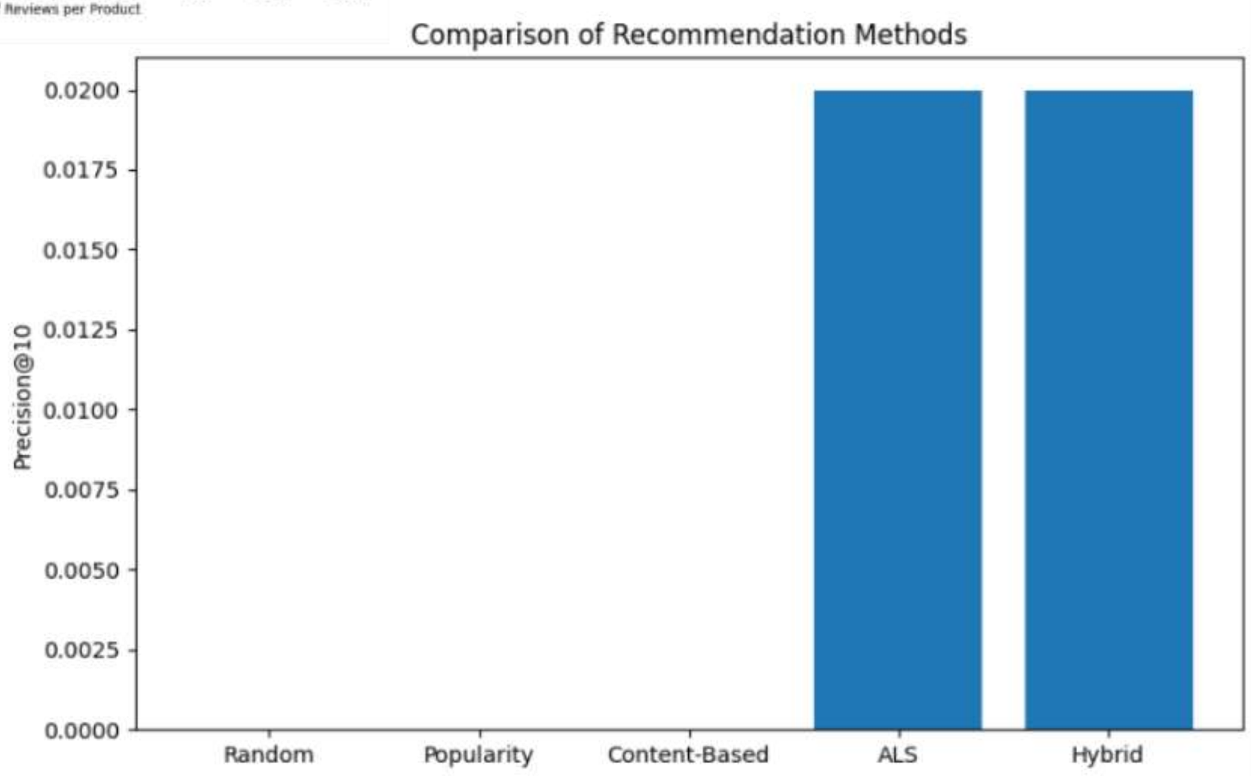
- Analyse premium negative reviews for detailed product feedback
- Target customer for specific season
- Marketing strategies : win-back for Long-Recency Customer

TASK 4 RECOMMENDATION SYSTEMS









CONCLUSION & FUTURE WORK

- Findings:
- Mongo – Qdrant – Bi-Secting K-means – Hybrid Recommender

- Use more databases
- Improve vector embeddings
- Experiment with deep learning recommenders
- Test on larger datasets